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Study on the Impact of Fuel Staging on Flame Structure and Thermoacoustic Oscillation in a Diesel Multi-Nozzle Combustor

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Modern gas turbine combustors usually adopt multinozzle lean premixed and pre-evaporated (LPP) combustion technology to reduce emissions. However, the thermoacoustic oscillation of this design poses a significant threat to the safe operation of gas turbines. This paper conducts experiments and numerical calculation studies on a diesel multinozzle LPP combustor under atmospheric pressure with inlet temperature near 380 °C to investigate the influence of its fuel staging ratio (FSR) on the flame structure and thermoacoustic oscillation. In the experiment, the pressure fluctuation and the self-luminescence images of OH were measured simultaneously. The flow field and flame structure were obtained by RANS calculation. The results show that for the diesel multinozzle LPP combustor, increasing the FSR can suppress the thermoacoustic oscillation of the combustor, and the oscillation amplitude is reduced by up to 60%. When the pilot stage is not ignited, the oscillation mode is double-period limit cycle oscillation, while after the pilot stage is ignited, the oscillation is single-period limit loop oscillation. The proper orthogonal decomposition mode shows that the main energy of the oscillation is concentrated in the global axial synchronous oscillation. The Rayleigh index and numerical calculation results indicate that the change of FSR alters the flame distribution in the combustor. The increase of FSR causes the flame to develop from the interference zone between the wall and the main stage to that between the main stage and the pilot stage. The change of flame structure is the main reason for the suppression of thermoacoustic oscillation.* [DOI: 10.1115/1.4069916]

Keywords: multi-nozzle, fuel staging, diesel combustion, thermoacoustic

With the continuous increase in environmental protection standard, low emissions have become an important technical indicator for gas turbines. Diesel, which is a common, safe and economic fuel of gas turbines has high energy capacity so suppressing NOx emission during combustion of diesel became a new topic of advanced gas turbine research. One of the emission control methods widely used to date is multinozzle LPP (Lean Premixed Prevaporized) combustion. Fuel is staged in the combustor and independently controlled based on the load required so thermal load is evenly distributed among nozzles in order to lower the emission. As an example, DLN2.6+ system from GE realized a low NOx emission of 5 ppm@15%O_2 [1]. Nonetheless, the thermoacoustic instability problem of multinozzle LPP combustion is more complicated than traditional single nozzle combustors [2,3] and poses great threat to the safe operation of gas turbines.

Lieuwen and Yang [4] pointed out that the triggering mechanism of oscillation combustion inside combustors is usually connected with multiple physical and chemical processes such as sound wave, fluctuation in flow, equivalence ratio or temperature of premixed gas, interaction between flame and turbulent flow or flame and sound wave, variance in flame front as well as fluctuation in heat release ratio. There are a lot of research on the response of single nozzle which provided useful insights into the cause of fluctuation in heat release rate including fluctuation in speed, swirling flow or equivalence ratio, interaction between flames and the stretching instability of flame [5–7]. Yu et al. [8] analyzed the thermoacoustic instability of laminar lean premixed diesel flame under self-ignition and conducted numerical simulation. The results showed that the overall heat release rate fluctuated with the flame front. The flame may amplify, attenuate, or filter acoustic energy depending on phase interactions between the velocity and pressure oscillations. Han et al. [9] studied the impact of air inlet temperature on thermoacoustic oscillation properties of LPP combustors and found that as temperature changes supercritical bifurcation occurred in the combustor and the amplitude of thermoacoustic oscillation changed abruptly when inlet temperature reaches 570 K. Biswas [10] researched the influence of variation in equivalence ratio on combustion instability mode and found that a decrease in equivalence ratio resulted in a drop in oscillation modes.

In the field of multinozzle combustor, KIM [11] looked into the effect of radical fuel staging on the mode of thermoacoustic instability, including varying fuel flowrate and different staging strategies. A method to identify axial and circular thermoacoustic coupled oscillation is developed base on experiments and CFD analysis [12]. Temme [13] conducted PIV on the oscillation

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characteristics of the LPP combustors and found that the oscillation originated from a repeated fluctuation of the base of flame between blowout and backfire. Liu [14] studied the suppressing effect of fuel staging in multinozzle array combustor on thermoacoustic instability. The results showed that when only the pilot nozzle was operational, ignition stability was ensured by local flue gas recirculation zones and a large recirculation zone in the middle of the combustor. Chen [15] studied how transient changes in air-fuel ratio affects thermoacoustic instability characteristics of single-nozzle and multinozzle combustors. It is found that the transient changes in air-fuel ratio affected the heat release rate distribution and flame shear layer near corner reticulation zones. Boundary restriction and flame interaction plays a vital role in flame dynamics.

Taesong Lee [16] studied the effect of fire interaction in two-nozzle combustors on the stability of combustion and the limit cycle phenomenon in multinozzle systems under nonsymmetrical flow responses and found that a nonsymmetrical flame could significantly suppress periodic oscillations. The convection time of vortex disturbance controlled by nonsymmetrical nozzles is systematically reviewed under a wide range of parameters. The results showed that self-oscillation is strongly connected to the arrangement of swirlers. The influence of flame interactions on FTF/FDF is also studied [17]. The limit cycle amplitude when there is flame interaction present is tightly connected to the extent of coupling strength of two heat sources. The combined effect of flame interactions and flame-sound interactions lead to an increase in combustion instability.

As discussed above, the way multinozzle combustor differs from the single-nozzle combustor is that the interference between adjacent flames is one of the main reasons affecting thermoacoustic oscillations. Therefore, the flame structure of the multinozzle combustor is also an important subject of research. The research by Hardalupas et al. [18] has shown that the chemiluminescence intensity of OH* and CH* is proportional to the equivalence ratio, and both can be used to describe the intensity of heat release rate in flames. Chong et al. [19] studied the spectra of diesel and biodiesel flames and found that the chemiluminescence of OH* and CH* in the flames of such fuels both have distinct radiation peaks. Moreover, Chong et al. [20] used the chemiluminescence intensity of CH* to investigate the flame structure of biodiesel. Sung et al. [21], Wei et al. [22], and Pickett et al. [23] measured and analyzed the flame structure of diesel combustion using OH* chemiluminescence. The images of OH* chemiluminescence intensity coincide with the images of soot incandescence, indicating that it is feasible to use OH* chemiluminescence to describe the flame structure.

There are many studies on the thermoacoustic oscillations of multinozzle combustors, but the fuel types are mostly limited to natural gas, hydrogen, and aviation kerosene. There are relatively few studies on diesel multinozzle LPP combustion. This research is based on a diesel 10-nozzle LPP combustor. Experimental and numerical simulation methods are used to study the impact of fuel staging on combustion stability. How changes in flame structure affect the thermoacoustic oscillations of diesel LPP combustion is also analyzed.

1 Research Methodology

1.1 Experiment System. The 10-nozzle diesel combustor test system consists of a combustor, air system, fuel system, cooling water system and purge air system as shown in Fig. 1.

The combustor is cylindrical with a length of 210 mm and an inner diameter of 165 mm, and the side walls have viewing windows for monitoring the combustion status.

The air system uses a 200 kW fan as the air source. The fan speed is controlled by a frequency converter to control the air flow. The air flow is measured by a vortex flowmeter. An electric heater preheats the air to a set temperature. The air then enters the combustor for combustion.

The fuel for the pilot stage and the main stage is controlled by two separate fuel systems. The fuel for the main stage is further divided into two branches. The first branch connects to six main combustion nozzles, while the second branch connects to three main combustion nozzles. The fuel pressure for all the main combustion nozzles is maintained the same.

The cooling water system consists of an exhaust section wall cooling system and a water spray cooling system. The wall cooling system delivers cooling water to the water-cooled jackets of the exhaust section through a water pump to cool the exhaust pipe. The water spray cooling system directly sprays cooling water into the exhaust pipe, where the evaporation of water droplets provides direct cooling of the high-temperature exhaust gas.

The purge air system is supplied by an air compressor. After each test session, compressed air is pumped into the fuel nozzles to blow out the residual fuel in the pipes and nozzles, preventing nozzle blockage caused by diesel coking. The ten-nozzle diesel combustor

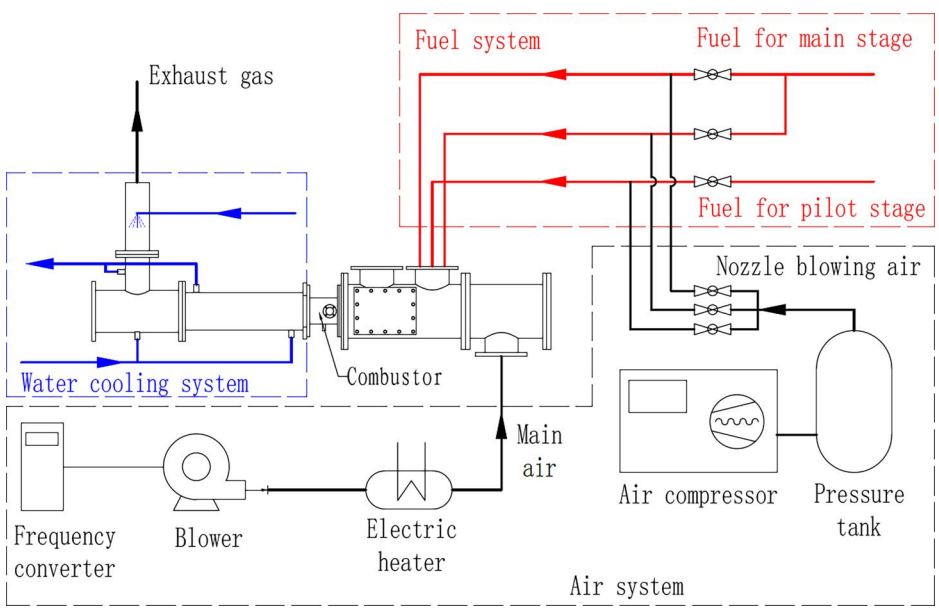

Fig. 1 Diesel 10-nozzle combustor experimental system

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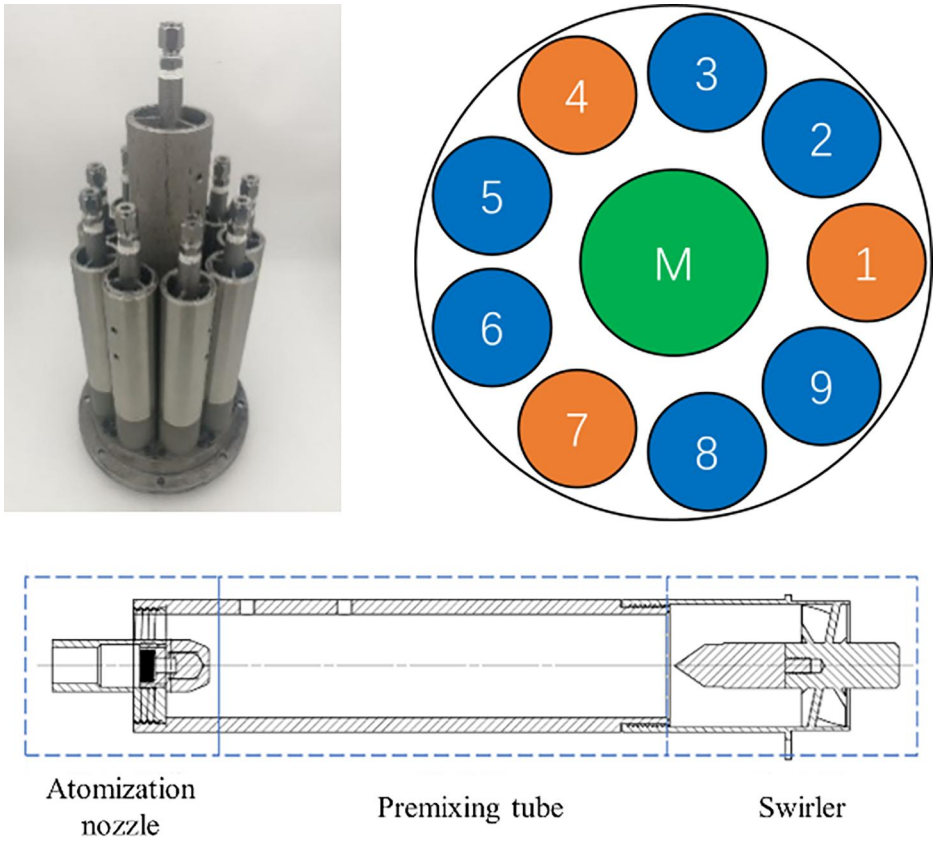

Fig. 2 Diesel 10-nozzle LPP combustor and nozzle serial numbers

and its structural diagram are shown in Fig. 2. It consists of one pilot nozzle (labelled M) at the middle and nine main nozzles (labelled as 1–9) in the outer ring. The nine main combustion nozzles are controlled in two groups. The first group consists of six main nozzles, while the second group consists of three nozzles. When ignited, the flowrate and equivalence ratio for all nine nozzles in the outer ring are uniform, but the fuel supply is controlled through two separate valves in a staged manner.

Each nozzle consists of a diesel spray nozzle, a premixing tube, and a swirler. The fuel is atomized by the diesel spray nozzle and then fully premixed and pre-evaporated in the preheated air. After passing through the swirler, it enters the combustor. The swirl number of the pilot nozzle swirler is 0.72, and the swirl number of the main combustion nozzle swirler is 0.61.

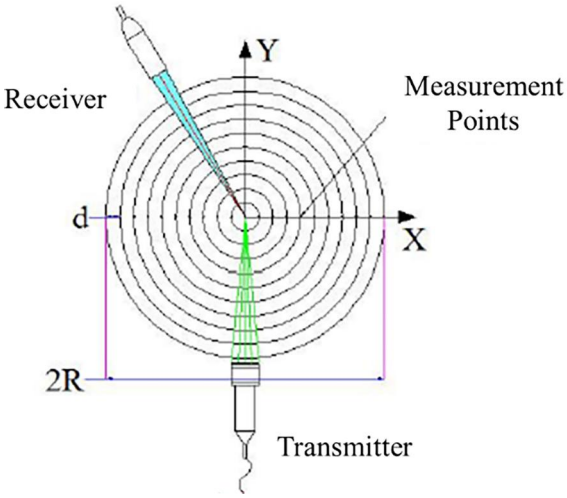

Fig. 3 Phase doppler particle analyzer measurement system

The air flow rates of the ten nozzles are freely distributed. According to experimental results, the air flowrate of the pilot stage accounts for approximately 19% of the total air flowrate, while the remaining 71% is occupied by the main stages. Therefore, the flowrate of one main stage is about 7.9%. The pressure drop of the burner is about 1.5%. The residence time is about 10.3 ms.

To ensure that the oil can completely evaporate in the premixing tube, the Sauter mean diameter (SMD, D_{32}) of the droplet was measured. The phase Doppler particle analyzer (PDPA) was used to measure the spray field, as shown in Fig. 3.

The SMD of droplets were measured under the oil pressure of 0.8 MPa, 1.4 MPa, and 1.95 MPa. At a distance of 15 mm from the nozzle outlet, the diameters of droplet particles were measured at equal intervals along the X-axis at several positions. The results are shown in Fig. 4.

It can be seen that the droplet diameter is less than 35 μm. Here, we can ignore the droplet fragmentation caused by the flow and calculate the theoretical evaporation time of the droplets under the experimental condition of 380 °C. The relationship between the droplet lifetime τ_d and the initial diameter D_0 is shown as follow:

$$\tau_d = D_0^2/K \tag{1}$$

where K is the evaporation constant. It satisfies the following formula:

$$K = \frac{8k_g}{\rho_l c_{p,g}} \ln(1 + B_q) \tag{2}$$

where $c_{p,g}$ represents the specific heat capacity at constant pressure of the gaseous phase. ρ_l is the density of the liquid phase. k_g represents the thermal conductivity coefficient of the gaseous phase. B_q is the dimensionless mass transfer number which satisfies the following formula:

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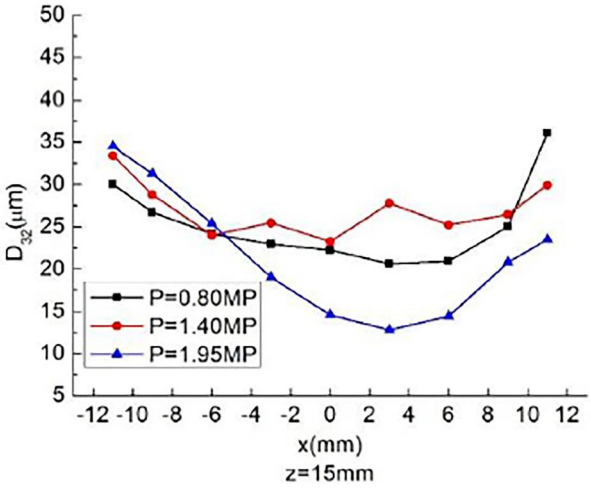

Fig. 4 The SMD distribution of droplets under different oil pressures

$$B_q = \frac{c_{p,g}(T_\infty - T_{\text{boil}})}{h_{fg}} \tag{3}$$

where T_∞ represents the temperature of the main flow, while T_{boil} represents the boiling point of the liquid phase. h_{fg} is the latent heat of vaporization.

In this paper, the physical property parameters of n-dodecane are used to estimate the physical properties of diesel. From this, the evaporation time of the droplets can be calculated. The evaporation time is about 0.0062 s. The average flow velocity of the gas in the pipeline is 19 m/s. That is to , say, the distance required for the droplet to completely evaporate is approximately 117 mm, while the actual premixing section length is 195 mm, which is long enough for the droplets to completely evaporate.

1.2 Measurement System. The measurement system of the ten-nozzle diesel combustor is shown in Fig. 5. This system enables the measurement of pressure and flame image. The pressure fluctuations within the combustor are measured using a Kulite pressure transducer, model XTL-190M-120KPaA, with a sensitivity of 0.84 mV/kPa. The transducer is 293 mm away from the head of the combustor with water cooling. Since the cross-sectional area of the combustor is much larger than that of the burner outlet, the inlet of the combustor is approximately a hard boundary condition in terms of acoustics, located at the wave trough, with the maximum amplitude. However, if the transducer is too close to the high-temperature zone at the combustor, it may be damaged. Therefore, this measurement position is much more suitable. The signal from the pressure transducer is synchronously acquired by a data acquisition instrument. The voltage resolution of the collector instrument is 0.1 mV. According to the sensitivity of the pressure transducer (0.84 mV/kPa), the error is within ±0.12 kPa.

An image intensifier is placed at the end of the test bench facing a viewing window and connected to a high-speed camera. A narrowband filter with a wavelength of 307 ± 13 nm is installed in front of the lens to obtain the self-luminous images of OH* in the flame. The images captured by the camera are sent to a computer. The exposure time of the camera is 495 μs. The lens aperture is f2.8. The camera is triggered by a mechanical switch. Aside from the camera I/O the trigger signal is also sent to the data acquisition instrument for synchronous collection of the pressure data. The camera frame rate is 1000 fps, and the data acquisition instrument has a sampling rate of 10,000 Hz. By comparing the trigger time of the camera, the flame images and pressure signals can be synchronized within an error of 0.1 ms.

1.3 Data Processing Method. This experiment adopted phase space reconstruction to study the thermoacoustic oscillation and its development patterns. Time delays of t_d and $2\,t_d$ were added to the signal $p(t)$ and the phase space diagram $[p(t),p(t+t_d),p(t+2t_d)]$ is plotted. The autocorrelation function $R(t_d)$ of the pressure signal was

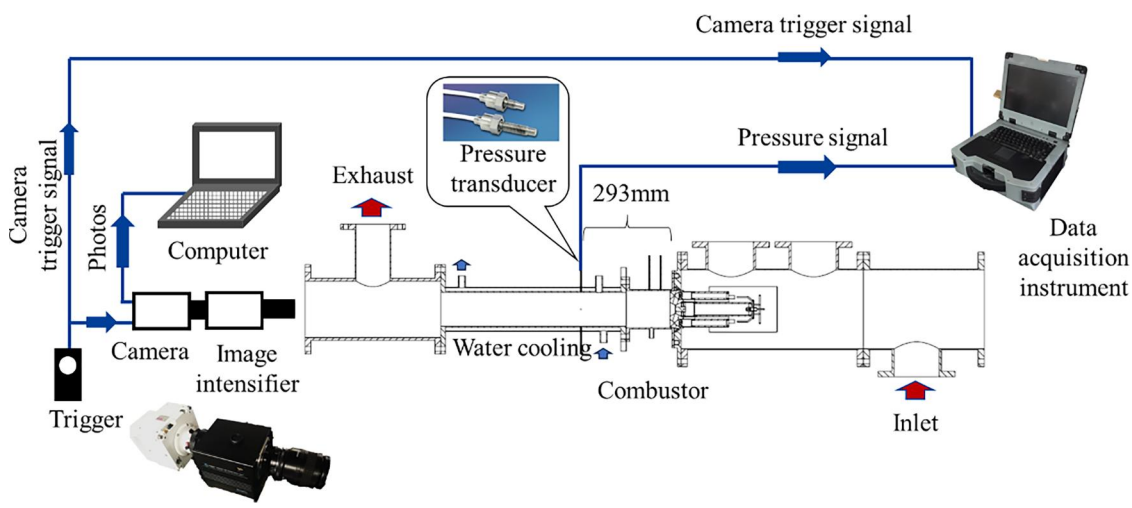

Fig. 5 Diesel 10-nozzle LPP combustor measurement system

Table 1 Experimental conditions of fuel staging

Case	Inlet temperature/°C	Air flow rate of STP/m^3 × h^{-1}	Equivalence ratio of the pilot stage	Equivalence ratio of the main stage	Total equivalence ratio	FSR
B1	372.1	399.3	0.00	0.67	0.55	0.00
B2	375.6	397.4	0.40	0.58	0.54	0.14
B3	376.2	398.9	0.49	0.55	0.54	0.17
B4	379.1	402.0	0.60	0.52	0.54	0.21
B5	388.9	399.9	0.67	0.51	0.54	0.23

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calculated using Eq. (4). When the value of $R(t_d)$ is less than 1×10^{-1} times the initial value $R(0)$, the time delay t_d is obtained

$$R(t_d) = \frac{1}{T}\int_0^T p(t)p(t+t_d)\mathrm{d}t \tag{4}$$

When the system is in a chaotic state without oscillation, the trajectory of the signal will converge to a core, known as an attractor. When the system's oscillation reaches its maximum, the trajectory of the signal will converge to a ring, known as a limit cycle. By analyzing the characteristics of the phase diagram, it is possible to determine whether the combustor has reached a limit cycle oscillation state.

This paper employs the method of proper orthogonal decomposition (POD) modal analysis to investigate the images of flame heat release rate. The POD modes can reflect the structural characteristics of the dominant energy in flame fluctuations. The method of POD modal analysis refers to the research of Sirovich [24] and Johnson et al. [25].

The POD method decomposes the flame heat release rate images into the form of superimposed orthogonal data modes as shown in the following equation:

$$q(x,t) = \sum_{i=0}^{M} \alpha_i(t)\varphi(x) \tag{5}$$

where x is the spatial coordinate, t is the temporal coordinate, α_i is the temporal coefficient of mode i, φ_i is the spatial distribution characteristic of mode i, and M represents the number of modes.

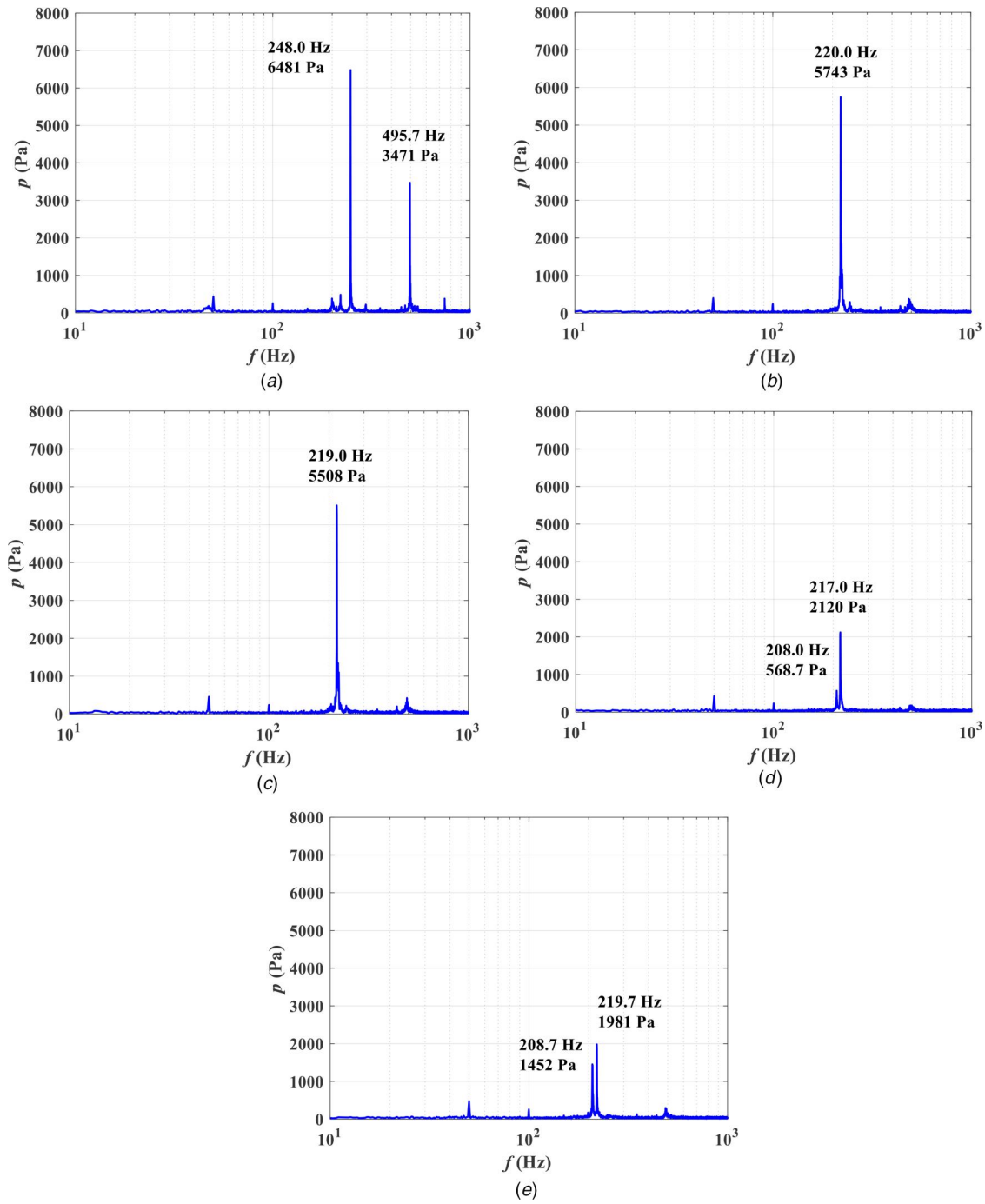

Fig. 6 Pressure fluctuation frequencies under different FSR

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Each image captured under a set of conditions is converted into a one-dimensional matrix a_i, and each set of operating conditions corresponds to a matrix A

$$A = (a_1^T, a_2^T, a_3^T, \ldots, a_n^T) \tag{6}$$

where n is the number of images. Singular Value Decomposition (SVD) of matrix A yields

$$A = U\Lambda V^T \tag{7}$$

where

$$\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n) \tag{8}$$

$$U = (u_1, u_2, u_3, \ldots, u_n) \tag{9}$$

$$V = (v_1, v_2, v_3, \ldots, v_n) \tag{10}$$

Here, Λ is the eigenvalue diagonal matrix, U is the spatial mode matrix, and V is the temporal mode matrix. λ^2 represents the energy value of each mode.

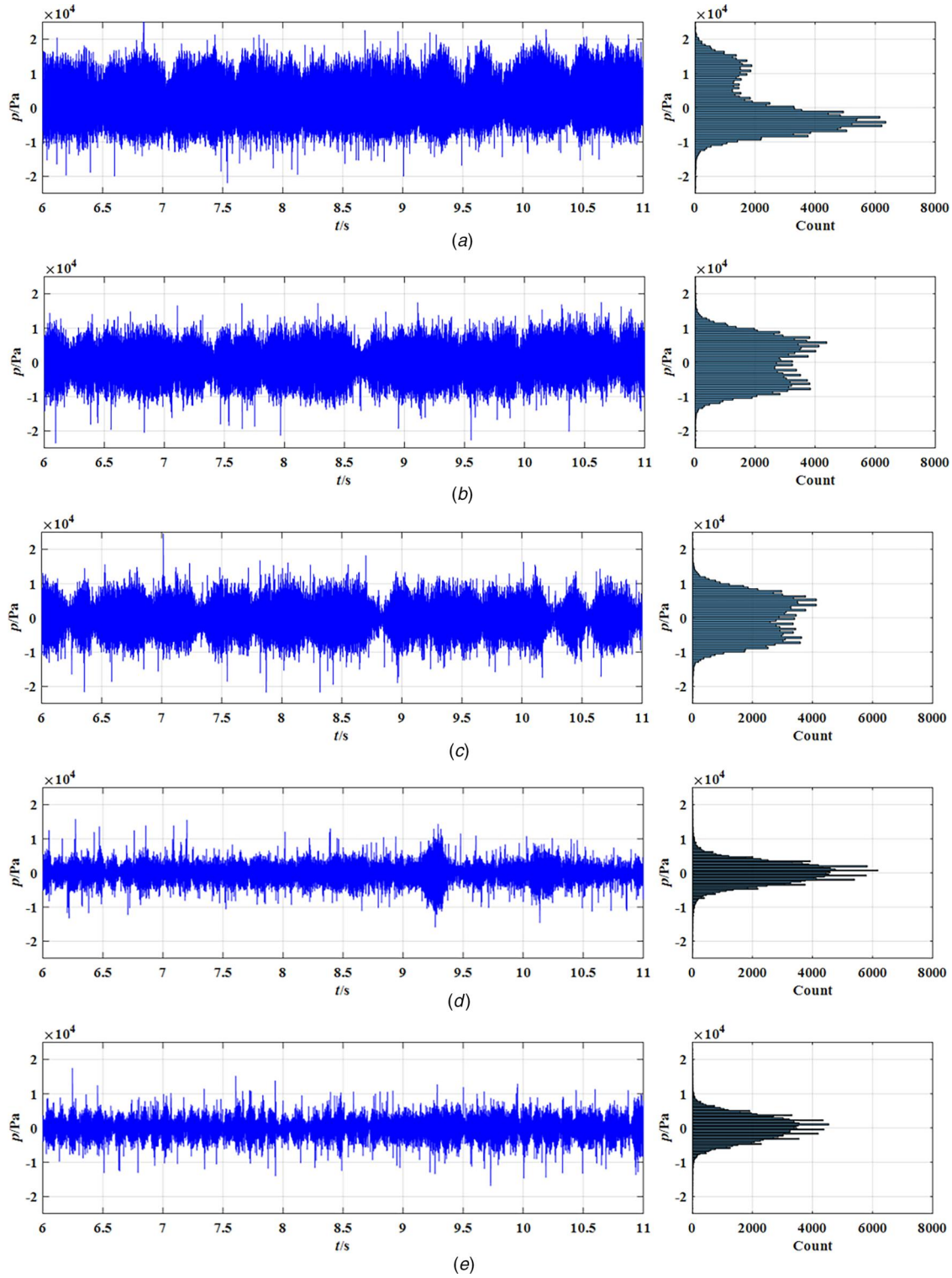

Fig. 7 Time domain signal of pressure fluctuation under different FSR

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In SVD, the eigenvalues in the obtained eigenvalue matrix are arranged in descending order. Therefore, the overall fluctuation characteristics of the flame can be represented by the first few modes.

To investigate the main locations where thermoacoustic oscillation is generated in the combustor, this paper also conducted an analysis of the Rayleigh Index (*RI*) using synchronized pressure signals and flame heat release rate images. The definition of the Rayleigh index is shown in the following equation:

$$RI = \frac{1}{T}\int_0^T p(t)q(t)\mathrm{d}t \tag{11}$$

When the phase difference between the pressure signal p and the heat release rate image q is less than 90 deg, RI is positive; otherwise, RI is negative. A positive RI indicates that the pressure and heat release rate at this location form a positive feedback loop, where fluctuations in the heat release rate lead to enhanced pressure fluctuations, thereby forming thermoacoustic oscillations. The Rayleigh Index can reflect the main locations where thermoacoustic oscillations occur.

1.4 The Experimental Conditions. This study investigates the influence of fuel staging on thermoacoustic oscillations. The experimental conditions are shown in Table 1. The air flowrate of standard temperature and pressure (STP, $T = 0\,^\circ\mathrm{C}, p = 1\,\mathrm{atm}$) and the total equivalence ratio kept constant, the fuel distribution ratio of the pilot stage and the main stage which is defined as the fuel staging ratio (FSR) is changed to study the changes in thermoacoustic oscillations. The FSR is defined as the ratio of the fuel flowrate of the pilot stage G_{pilot} to the total fuel flowrate G_{total}.

$$\mathrm{FSR} = G_{\mathrm{pilot}}/G_{\mathrm{total}} \tag{12}$$

The flowrate control accuracy of the fuel is 0.1 kg/h ($\pm 1\%$), while that of the air is $0.1\mathrm{m}^3/\mathrm{h}$ ($\pm 0.1\%$). Therefore, the error in the equivalent ratio should be $\pm 1.1\%$, and the error in the FSR should be $\pm 2.2\%$.

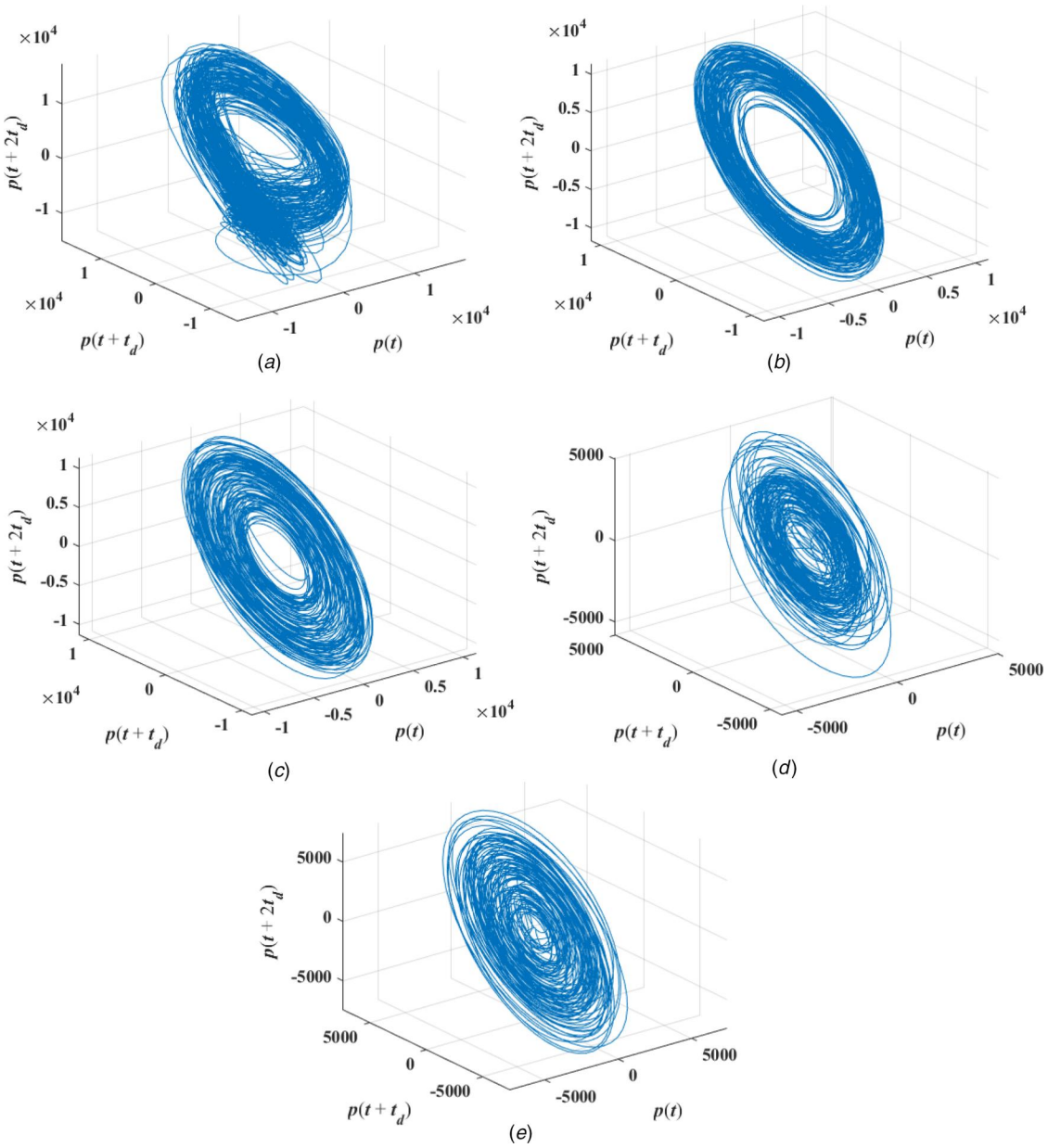

Fig. 8 Pressure fluctuation phase space diagram under different FSR

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1.5 Numerical Simulation Method. To further investigate the relationship between thermoacoustic oscillations and flame structure, numerical calculations were also conducted on the basis of experiments in this paper. The flow field and OH radical distribution of the multinozzle combustor under different FSR were calculated using the Reynolds-Averaged Navier–Stokes (RANS) method. The turbulence model used in the calculation is the Realizable k-ε model. This model is chosen because it is more accurate in simulating swirling flow fields. According to the introduction in the Fluent user manual, the Realizable k-ε model is also known as the k-ε model with swirl correction. Compared to the standard k-ε model, it performs well in swirling flows, strong reverse pressure gradients at boundary flows, flow separation, and secondary flows. The flamelet generated manifolds, FGM, is also used. The reaction mechanism used refers to the research of Luo et al. [26]. In their study, the 105-species skeletal mechanism of *n*-dodecane can well simulate the

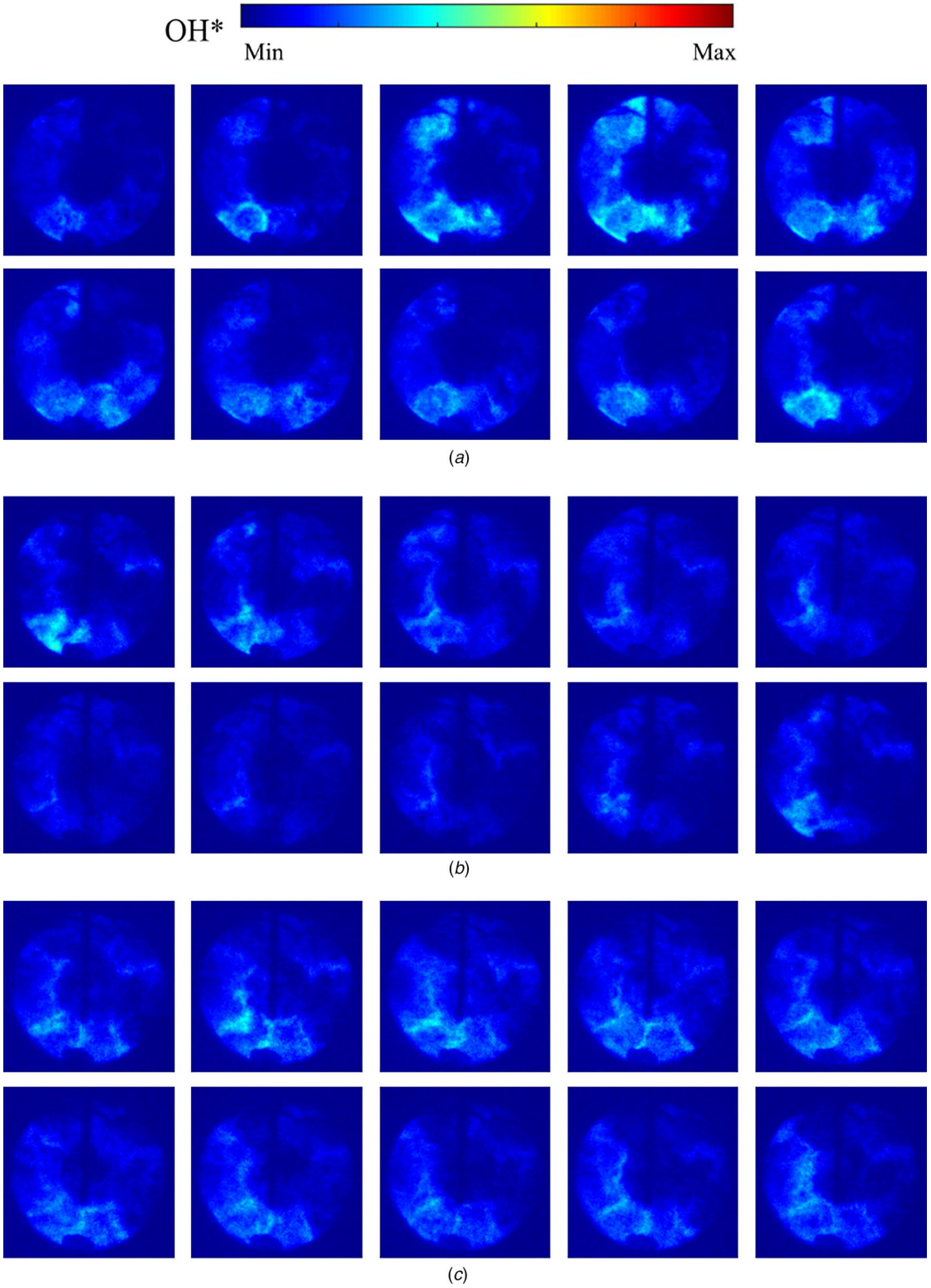

Fig. 9 OH* self-luminous images under different FSR

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reaction kinematics of diesel. The same mechanism was also adopted in the calculations of this paper. The inlet is set as a mass flow boundary, and the outlet is set as a pressure boundary. The grid used has 770,000 cells. The calculation conditions are the same as the experimental conditions.

In the calculation, thermal losses or conjugate heat transfer are not modeled. Instead, isothermal boundary condition with a temperature of 800 °C is adopted on the wall boundary conditions of the combustor. For the wall upstream, an isolated boundary condition is used.

2 Results and Discussion

2.1 Effect of Fuel Staging on Pressure Fluctuation Characteristics. The pressure fluctuation signals of the combustor under different FSRs were analyzed using Fast Fourier Transform (FFT) with a sampling rate of 10,000 Hz. The analysis was conducted based on 30,000 data points, resulting in a frequency resolution of 0.33 Hz. The data overlap rate was 70%, and the results were averaged after repeating the analysis 15 times, yielding the pressure fluctuation spectrum shown in Fig. 6. Here, f represents frequency, and FSR denotes the fuel staging ratio.

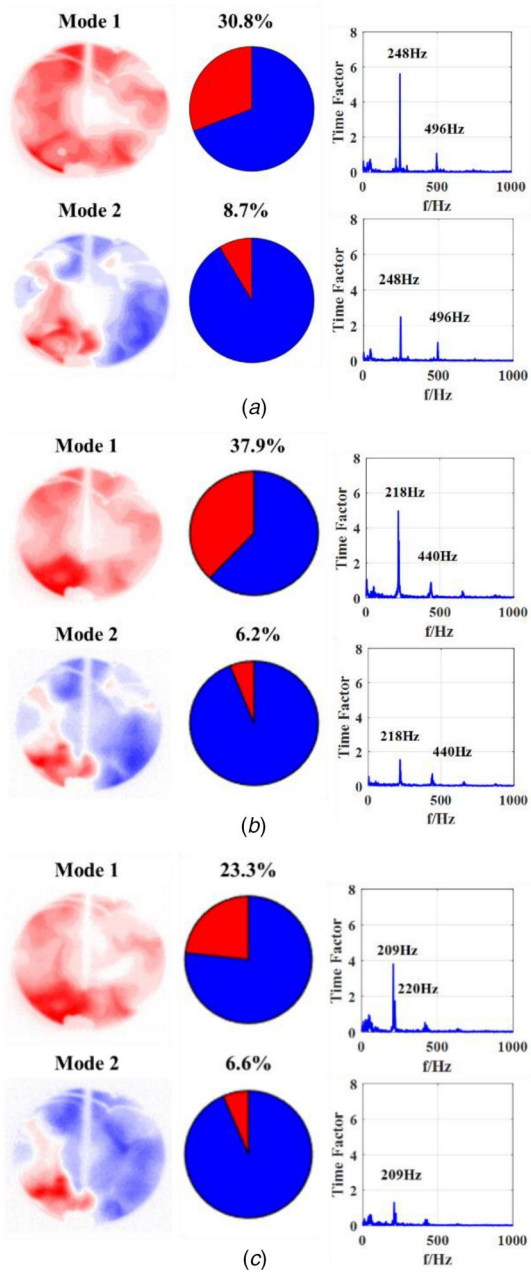

Fig. 10 First three orders of POD modes of flame images and their respective energy percentage under FSR of 0.00

When the FSR was 0.00, meaning the pilot stage fuel flow was zero, the combustor pressure fluctuation had two dominant frequencies: 248.0 Hz and 495.7 Hz. The 495.7 Hz frequency was a second harmonic of the 248.0 Hz frequency.

After the pilot stage was activated, the dominant oscillation frequency decreased by 11% to 220.0 Hz. The amplitude also dropped by 11%, and the harmonic virtually disappeared. This change was due to the alteration of the flame structure within the combustor caused by the introduction of the pilot-stage fuel, leading to a sudden shift in the dominant oscillation frequency.

As the FSR continues to increase, the amplitude of the combustor pressure fluctuation decreases, with the frequency ranging between 217.0 Hz and 220.0 Hz. The maximum amplitude is 5743 Pa, while the minimum is 1981 Pa, indicating a reduction ratio of the fluctuation amplitude up to 65%. Moreover, when the FSR is increased to 0.21, a peak around 208 Hz appears, and its amplitude increases with the rise of the FSR.

The time-domain signals of combustor pressure fluctuation and their statistical histograms under different FSRs are shown in Fig. 7. When the FSR is 0.00, the statistical histogram of pressure fluctuations exhibits an asymmetric bimodal distribution. This is the result of the combined effect of the two dominant oscillation frequencies, which have similar phases. This causes the oscillation at 247.7 Hz, when superimposed with the harmonic at 495.3 Hz, to strengthen one peak and weaken the other within a single cycle, thus forming an asymmetric structure.

When the FSR is increased to 0.14 and 0.17, the statistical histograms of pressure fluctuations show a bimodal structure. At this point, the pressure fluctuations enter a stable periodic oscillation state. As the FSR further increases to 0.21 and 0.23, the statistical histograms of pressure fluctuations revert to a unimodal form and approach the characteristics of a Gaussian distribution. This indicates that the oscillations are evolving toward a stochastic oscillation mode.

Phase space reconstruction was performed on the pressure fluctuation signals under different FSRs, resulting in the phase space diagrams shown in Fig. 8. Here, t_d represents the time delay (the calculation method is described in Sec. 1.3). When the FSR is 0.00, the phase diagram of pressure fluctuations shows a distinct double-period limit cycle structure, indicating that the combustor is undergoing limit cycle oscillations. This double-period structure is formed by the combined effect of the dominant frequency of the pressure fluctuations and its harmonics.

When the FSR is greater than 0.21, the limit cycle structure in the phase diagram becomes unstable, showing a tendency to converge toward a central attractor. The combustor pressure fluctuations decrease and tend toward a stochastic oscillation mode.

These results indicate that adjusting the FSR can suppress the thermoacoustic oscillations in the combustor.

2.2 Effect of Fuel Staging on Heat Release Rate Fluctuation. In addition to measuring the combustor pressure fluctuations, this paper also captured flame images under fuel staging conditions using a high-speed camera. By employing a filter, the OH* chemiluminescence distribution in the combustor was obtained whose intensity represents the intensity of the flame heat release rate.

Flame images under different Fuel Staging Ratios (FSR) are shown in Fig. 9, with each image separated by a time interval of 0.5 ms. When the FSR is 0.00, the flame heat release rate intensity exhibits significant periodic oscillation characteristics. At an FSR of 0.17, the flame still has periodicity, but the oscillation intensity is weakened. When the FSR is 0.23, the oscillation intensity is further reduced, making it difficult to directly identify periodicity from the images. A total of 1,000 flame images were used for POD to obtain the first three POD modes under different FSRs. Their mode

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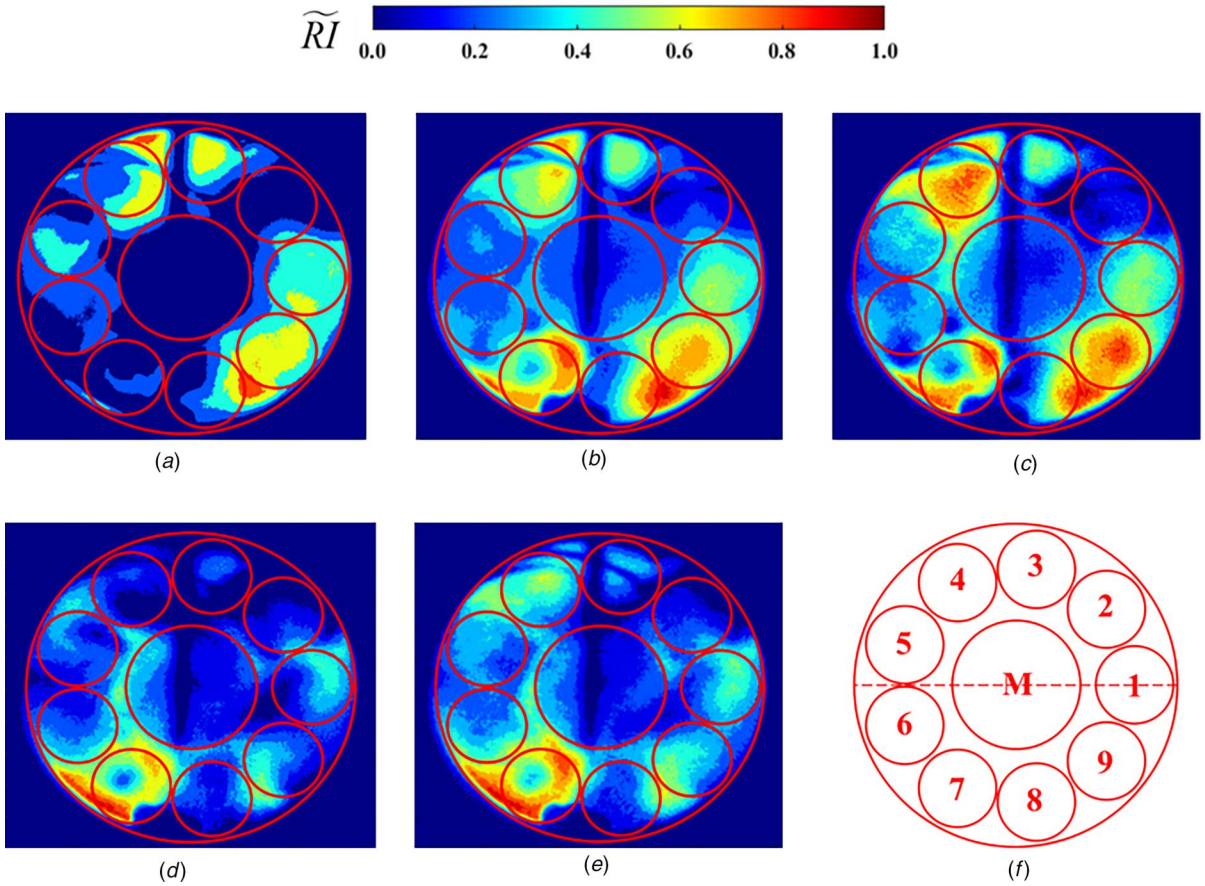

Fig. 11 The normalized Rayleigh index of the multinozzle combustor under different FSR

distribution maps and energy proportions are shown in Fig. 10. The spectrum in the figure represents the FFT results of the time coefficients of each mode. The colors in the mode maps represent different phases. It can be observed that the first mode has the largest energy proportion under different FSRs. The oscillation phases of the nozzles under the first mode are basically consistent. Since the high-speed images were captured through the tail window, this mode belongs to global axial synchronous oscillation. The second and third modes are local oscillations with similar energy proportions. Both have radial oscillation characteristics but in different directions. However, due to their lower energy proportions, they are not the main components of thermoacoustic oscillations. The spectrum of the time coefficients shows that the oscillation frequencies of the POD modes under different FSRs are consistent with the main frequencies of the pressure fluctuations.

To further investigate the coupling relationship between pressure and heat release rate, this paper conducted a Rayleigh index analysis. The normalized Rayleigh index $\tilde{RI} = RI/RI_{\max}$ distribution in the multinozzle combustor is shown in Fig. 11. In the normalized index analysis, red areas in the figure indicate regions with higher Rayleigh index values, where the thermoacoustic coupling is stronger, while blue areas indicate regions with lower Rayleigh index values, where the thermoacoustic coupling is weaker. The Rayleigh index reflects the regions where thermoacoustic oscillations are generated. It can be seen that when the pilot stage is not activated, i.e., FSR = 0.00, the Rayleigh index is concentrated in the interference zones between nozzles 3 and 4, and nozzles 8 and 9. The thermoacoustic oscillations are mainly caused by the interference between the main stages. When FSR = 0.14, a significant Rayleigh index is observed in the region between nozzle 7 and the pilot stage. As the FSR increases, the Rayleigh index in the interference zones between the pilot stage and the main stages rises. When FSR > 0.21, the Rayleigh index in the interference zones between the main stages decreases, indicating that the interference between the main stages contributes less to the thermoacoustic oscillations, and the main oscillations are concentrated around nozzle 7.

Figure 12 shows the Rayleigh index distribution in the range marked by the dashed line in Fig. 11(*f*). The black dashed line in the figure indicates the location of the pilot stage exit. The dip at the $X = 0$ position is due to the blockage by a thermocouple in the experiment. As can be seen from the figure, the Rayleigh index at different locations under different FSRs is generally greater than zero, indicating that the oscillations in the multinozzle combustor

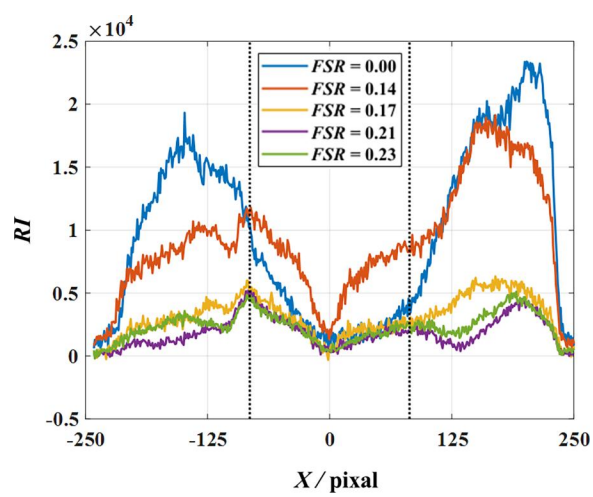

Fig. 12 The radial distribution of the Rayleigh index under different FSR

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are global. When FSR = 0, the RI is concentrated in the main stage area. After the pilot stage is activated, the RI in the interference zones between the main stage and the pilot stage increases, while the RI in the main stage decreases. As the FSR is further increased, the global RI decreases, and the RI in the interference zones between the pilot stage and the main stage gradually dominates. The spatial distribution changes of the RI may be due to the changes in the spatial distribution characteristics of the heat release rate.

2.3 Numerical Simulation of Multi-Nozzle Combustor Flow Field. However, due to the free distribution of air at the nozzle inlets of the test system discussed above and the large number of nozzles, it is difficult to ensure the uniformity of air distribution, which leads to the uneven spatial distribution of *RI* as shown in Fig. 11. Moreover, due to the blockage by a thermocouple, some images are not fully displayed. To further investigate the role of the flame heat release rate distribution in the thermoacoustic oscillations of the

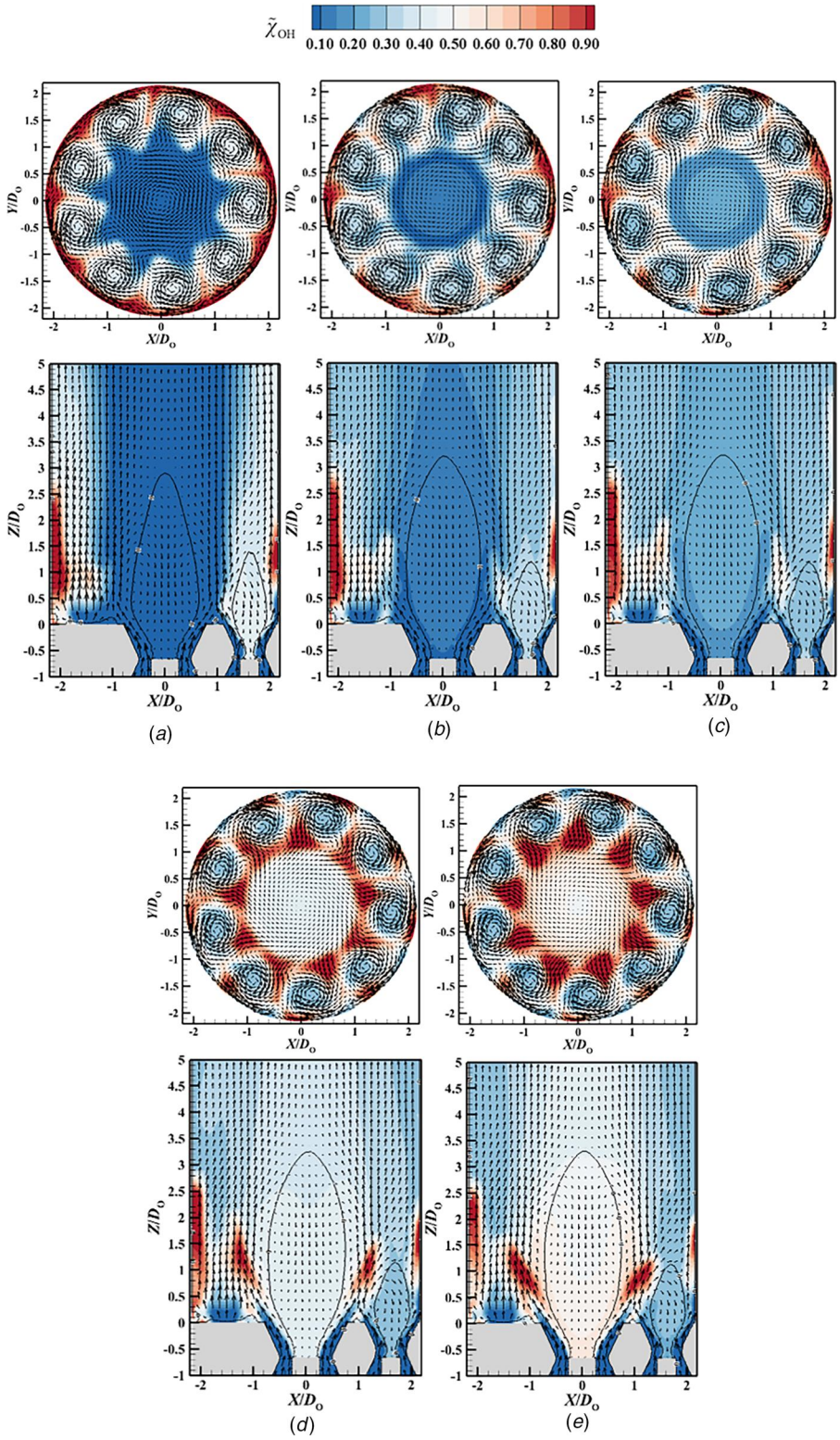

Fig. 13 Numerical calculation results of multinozzle combustion flow field and normalized OH radical mole fraction distribution

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multinozzle combustor, numerical simulations of the flow field in the multinozzle combustor were conducted. Figure 13 shows the flow field vector diagram and the spatial distribution of the OH radical in the multinozzle combustor. Since the OH radical mole fraction (χ_{OH}) varies greatly under different operating conditions, in order to make the image distinguishable, the OH radical mole fraction was normalized ($\tilde{\chi}_{OH}$). D_o represents the diameter of the main nozzle outlet, so both the X and Y axes represent the normalized scales. The color in the figure represents the magnitude of the normalized OH radical mole fraction, which is defined as in Eq. (8). The maximum OH radical mole fraction under the current operating condition is denoted by $\chi_{OH,max}$. Figure 14 shows the radial distribution of the un-normalized OH radical mole fraction.

$$\tilde{\chi}_{OH} = \chi_{OH}/\chi_{OH,max} \tag{13}$$

When only the main combustion nozzles are activated, i.e., FSR = 0.00, the OH radical is mainly concentrated in the wall region and the interference zone between the main combustion nozzles. This indicates that in the absence of pilot flame, the flame in the multinozzle combustor is determined by the interaction between the main combustion flow field and the combustor wall, and the flow interference between adjacent main combustion nozzles is also a major reaction zone.

When FSR = 0.14, flame is also generated in the flow interference region between the pilot and main stages. When FSR > 0.21, the OH radical mole fraction in the interference zone between the main combustion and pilot stages exceeds that in the wall region, and the interference flame between the pilot and main stages dominates. A triangular reaction zone is formed between every two main combustion nozzles and the pilot stage. As the FSR increases, the angle of the pilot-main combustion interference flame also increases significantly.

As shown in Fig. 14, the radial distribution characteristics of the OH radical mole fraction at the position of $Y = 0$, $Z = 1.0D_o$ are presented. It can be observed that when the pilot stage is not activated, there is no peak of OH radical between the pilot and main stages, and the main chemical reactions are concentrated near the combustor wall. After the pilot stage is activated, i.e., at FSR = 0.14, an interference reaction zone between the pilot and main stages appears at the position of $X = 1.0D_o$. As the FSR increases, this reaction zone gradually expands outward, and the OH radical mole fraction in this zone gradually increases, while the OH radical mole fraction near the wall decreases.

The black lines in the XZ plane images in Fig. 13 mark the recirculation zones in the combustor. Before the pilot stage is activated, i.e., at FSR = 0.00, the pilot stage recirculation zone is conical. After the pilot stage is activated, the pilot stage recirculation zone takes on a spindle shape. This is because, in the absence of the pilot flame, the temperature of the pilot stage gases is much lower than that of the main stage exhaust gases. The pilot stage recirculation zone is compressed by the expanding high-temperature exhaust gases from the main stage, forming a conical structure. After FSR > 0.14, the pilot stage gases also expand in volume, causing the recirculation zone to expand and take on a spindle shape.

Figures 15 and 16 show the lengths of the recirculation zones for the pilot and main stages, respectively. It can be observed that as the FSR increases, the pilot stage recirculation zone gradually lengthens, while the main stage recirculation zone shortens.

Based on the combined analysis of the experimental and numerical results, the significant changes in the chemical reaction zone are the main reasons for the suppression of thermoacoustic oscillations. In the absence of the pilot flame, the heat release rate is more concentrated, and the interactions between the main stages and between the main stages and the wall strengthen the instability of the flame. As the Fuel Staging Ratio (FSR) increases, the chemical reaction zone is divided into two parts: the wall region and the interference zone between the pilot and main stages. The local heat release rate intensity is weakened, and the oscillations are suppressed.

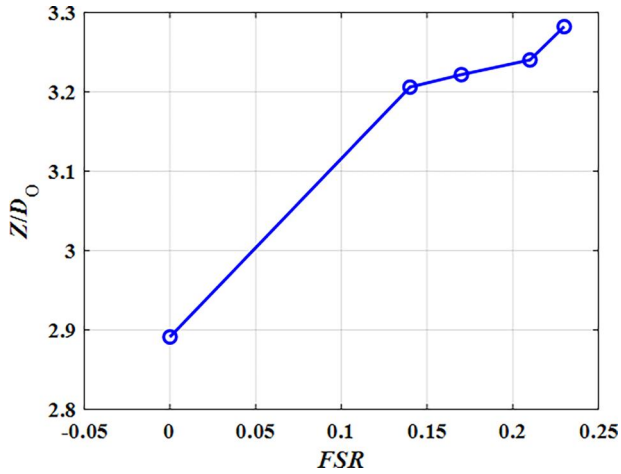

Fig. 15 The length of the recirculation zone of the pilot stage varies with FSR

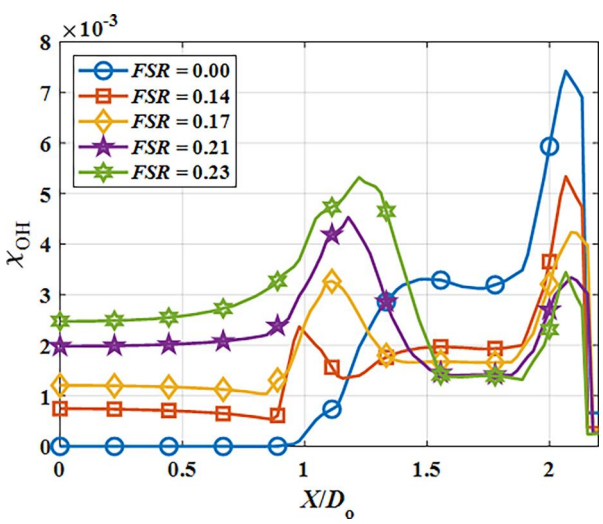

Fig. 14 The OH radical mole fraction distribution of multinozzle combustor

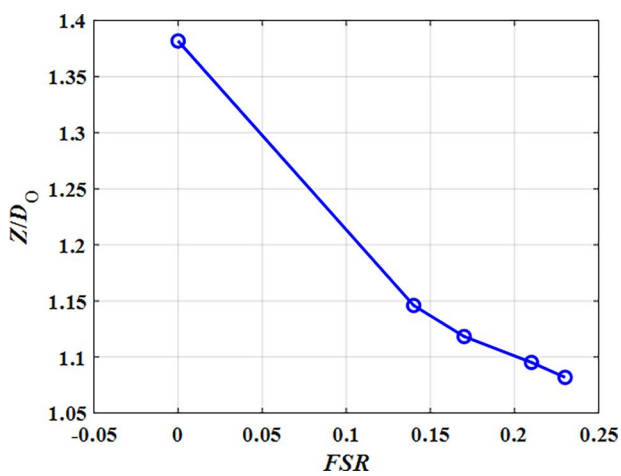

Fig. 16 The length of the recirculation zone of the main stage varies with FSR

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3 Conclusion

This paper studies the thermoacoustic oscillations in a multi-nozzle diesel LPP combustor under fuel staging conditions. Through the experimental method of OH* chemiluminescence image and pressure fluctuation signals synchronized acquisition, the causes of thermoacoustic oscillations were studied using POD modal analysis and Rayleigh index analysis. Numerical calculations were also conducted to obtain the distribution characteristics of the OH radical in the combustor. The main conclusions are as follows:

(1) For the multi-nozzle diesel LPP combustor, whether the pilot stage is activated or not has a significant impact on the oscillation frequency. The frequency is reduced by 11% when the pilot stage is activated compared to when it is not. Adjusting the fuel FSR can suppress the pressure fluctuations in the combustor, with the fluctuation amplitude reduction reaching up to 60%.

(2) Under the condition without the pilot stage, the oscillation mode is an asymmetric double-peak oscillation. The phase space image indicates that the oscillation is a double-period limit cycle oscillation. After the pilot stage is activated (FSR$\leq$0.17), the oscillation mode becomes a symmetric double-peak oscillation, and the phase space image shows that the oscillation is a single-period limit cycle oscillation. When FSR is greater than 0.21, the oscillation mode changes from double-peak to single-peak oscillation, and the phase diagram evolves from a single-period limit cycle toward chaos.

(3) The main energy of the oscillation is concentrated in the global axial synchronous oscillation. The Rayleigh index and numerical calculation results show that changes in FSR alter the flame distribution in the combustor. An increase in FSR causes the flame to shift from the wall and main stage interference zone to the main stage-pilot stage interference zone. This change in flame structure is the main reason for the suppression of thermoacoustic oscillations.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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